STIVEN CATALYST

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MANAGEMENT REVIEW

Management without theatre.



STRATEGY

THE BOTTLENECK: THEORY OF CONSTRAINTS

Herbie on the scout hike, what published TOC cases reported, the five focusing steps, what 42 healthcare projects found, three measures instead of efficiency, and a card for your own bottleneck.

69% was the average cut in lead time in 32 published applications of the theory of constraints. None of the reports the authors found described a failure.

BALDERSTONE & MABIN, 1998

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In this issue

Every operation has one step that sets the pace for all the others. Improve anything else and the output stays the same. This issue is about finding that step, working it to the full, and noticing when it moves.

In 1984 Eliyahu Goldratt and Jeff Cox told the theory of constraints as a novel, The Goal. Published applications reported average cuts in lead time of around 70%, though not one report of a failure turned up. The five focusing steps show the order of the work. In 42 healthcare projects, most reported gains, and in one the queue moved to the next ward. Three measures replace local efficiency.

Stiven Janaqi EDITOR · STIVEN CATALYST

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HOW TO READ THIS ISSUE

Figure
Every figure has its source and year at the foot of its page.

Our reading
Where the editors interpret rather than the research, it says so.

Practice
The steps and the card are proposals to try, not research results.

COVER STORY

Herbie sets the pace

In The Goal (1984), plant manager Alex Rogo leads his son's scout troop on a hike. The line stretches out. The boys cannot overtake the one in front, and the slowest of them, Herbie, decides how fast the whole troop moves.

01

Dependent events

each hiker can go only as fast as the one ahead allows

02

Fluctuations

the delays add up along the line; the gains do not make up for them

03

Bottleneck

in Goldratt's terms, any resource with less capacity than the demand on it

Alex puts Herbie at the front, and the boys share out his heavy pack. The troop stays together and walks faster. Goldratt wrote the novel for workers in plants using his scheduling software, OPT. Its schedules left some stations idle; measured on their own efficiency, workers kept busy and built stock.

OUR READING

Everyone busy is not the same as everything moving. The line goes at Herbie's pace, however fast the others walk.

Sources: Eliyahu M. Goldratt & Jeff Cox, North River Press, 1984 (via Watson, Blackstone & Gardiner, 2007; LitCharts); Kevin J. Watson, John H. Blackstone & Stanley C. Gardiner, Journal of Operations Management 25(2), 2007

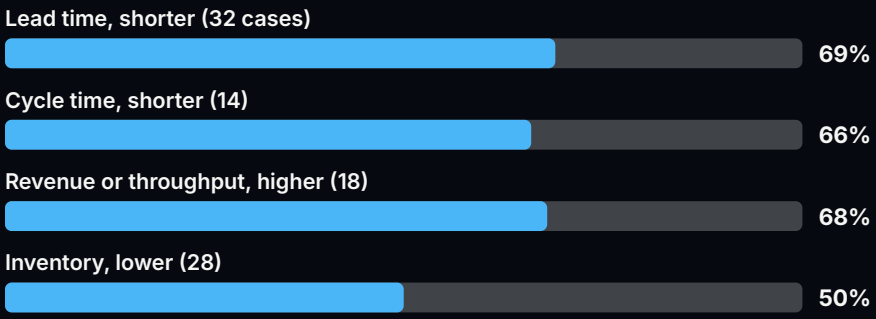
The hike is fiction from the novel; we know it from summaries, not from the book itself. Why the book was written follows Watson, Blackstone and Gardiner (2007).

THE NUMBERS

What the cases reported

Steven Balderstone and Victoria Mabin, at Victoria University of Wellington, searched the literature on the theory of constraints. Of over 100 case reports, 77 companies gave figures. The averages, with the number of cases behind each:

MEAN CHANGE REPORTED IN PUBLISHED TOC APPLICATIONS, 1998



In their 2003 article, with over 80 applications, the means were close: lead time 70% shorter, cycle time 65% shorter, inventory 49% lower. In both searches the authors found no report of a failure.

OUR READING

An average of published successes is not an average of attempts. It shows what is possible, not what to expect.

Sources: Steven J. Balderstone & Victoria J. Mabin, Victoria University of Wellington, conference paper, 1998; Victoria J. Mabin & Steven J. Balderstone, International Journal of Operations & Production Management 23(6), 2003 (via Panizzolo, 2016); Roberto Panizzolo, International Journal of Industrial Engineering and Management 7(1), 2016

Reports by the companies or their authors, mostly from manufacturing in North America; the revenue mean leaves out one rise of 600%. The 2003 means as a later study cites them.

THE MODEL

Five focusing steps

The Goal sets out the order in which the theory of constraints is put to work. Later accounts add two things to do first: define the system and its purpose, and choose measures that tie it to that purpose.

1. **Identify the constraint**
   the one step that limits the output of the whole
2. **Exploit it**
   get the most from it with the resources you already have
3. **Subordinate everything else**
   the other steps work at its pace, not at their own
4. **Elevate it**
   add capacity there, only if more output is still needed
5. **Do not let inertia set in**
   if the constraint has moved, start again

A constraint can be physical, when capacity is below demand; the market, when demand is below capacity; or a policy, a written or unwritten rule that holds the system back.

OUR READING

Exploit comes before elevate. The cheapest hour of capacity is the one the bottleneck loses today, waiting.

Sources: Eliyahu M. Goldratt & Jeff Cox, North River Press, 1984 (via Watson, Blackstone & Gardiner, 2007; LitCharts); Theory of Constraints Institute, 2021; Kevin J. Watson, John H. Blackstone & Stanley C. Gardiner, Journal of Operations Management 25(2), 2007

Steps after the Theory of Constraints Institute; the prerequisites and the three kinds of constraint after Watson et al. (2007), who cite the APICS Dictionary. The reading is the editors'.

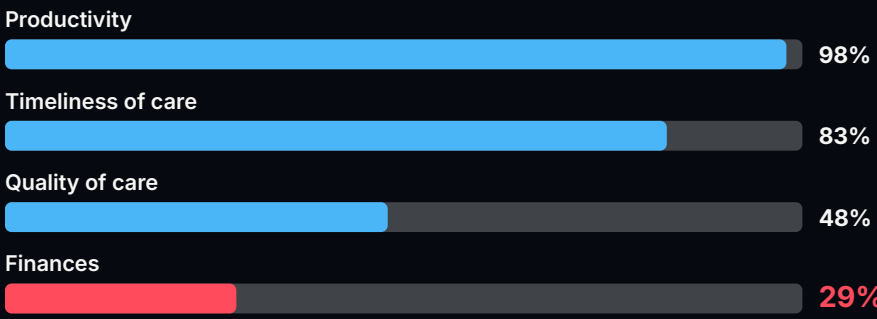
More in the essay High-volume days: the standard under pressure

WHAT THE RESEARCH SAYS

When the queue moves on

Gabriel Bacelar-Silva, James Cox and Pedro Rodrigues reviewed 42 uses of the theory of constraints in healthcare, from journals, theses and conference talks, up to 2020. The five focusing steps were the method used most, in 76%. Share of projects that reported a gain:

42 TOC PROJECTS IN HEALTHCARE, 2020



Where they were measured, waiting times fell by half on average. In three, one measure did not change. In one, more patients got through, and the ward and the follow-up care after it became congested.

OUR READING

Lift one bottleneck and the queue moves to the next. That is why the fifth step exists.

Source: Gabriel M. Bacelar-Silva, James F. Cox III & Pedro P. Rodrigues, Health Systems 11(1), 2020

The authors call their review exploratory: few projects, data often incomplete, no statistical tests, and most applied only parts of the method.

HOW IT IS MEASURED

Three measures, one pace

In place of the efficiency of each station, The Goal measures the plant as a whole with three numbers. Its aim: raise throughput while cutting inventory and operating expense.

01
Throughput
the rate at which the system makes money through sales

02
Inventory
the money tied up in things it intends to sell

03
Operating expense
the money spent to turn inventory into throughput

Keeping to the schedule is a fair measure at the constraint. Elsewhere, running at full output only builds stock.

HYPOTHETICAL EXAMPLE, AN EVENING SHIFT IN A DISPATCH AREA

Constraint	the loading ramp: 4 doors, 16 door-hours
Lost there	2.5 door-hours waiting for trolleys
Elsewhere	picking at 105% of plan, 30 trolleys queuing

Picking looks best and the ramp loses the most. The numbers are invented.

Sources: Eliyahu M. Goldratt & Jeff Cox, North River Press, 1984 (via Watson, Blackstone & Gardiner, 2007; LitCharts); Kevin J. Watson, John H. Blackstone & Stanley C. Gardiner, Journal of Operations Management 25(2), 2007

The three measures and their definitions are Goldratt's, as Watson et al. (2007) and a textbook glossary give them. The example is the editors'.

TOOL OF THE ISSUE

The bottleneck card

One process, one constraint. Look for where the work waits, not where people look busiest, and write down the date when you will check whether it has moved.

01 **Process and goal** where it starts and ends; what output counts

02 **1. The constraint** where work queues; physical, market or a rule

03 **2. Hours it loses** waiting, breaks, rework; what changes at no cost

04 **3. What the others change** release, pace and priorities set by the constraint

05 **4. Added capacity** only if still short after step 2; what it costs

06 **5. Check again** date; where the queue is now

Sources: Theory of Constraints Institute, 2021; Gabriel M. Bacelar-Silva, James F. Cox III & Pedro P. Rodrigues, Health Systems 11(1), 2020

A practice proposed by the editors, after the five focusing steps and the findings above.

Open the tool: Delay Analyzer stivencatalyst.com/tools/delay-analyzer/

SOURCES AND METHOD

Every figure has a source.

The figures in this issue come from the sources below. The year shows how recent each one is.

The Goal: A Process of Ongoing Improvement
Eliyahu M. Goldratt & Jeff Cox, North River Press, 1984 · via Watson, Blackstone & Gardiner, 2007; LitCharts

The evolution of a management philosophy: The theory of constraints
Kevin J. Watson, John H. Blackstone & Stanley C. Gardiner, Journal of Operations Management 25(2), 2007
https://doi.org/10.1016/j.jom.2006.04.004

A Review of Goldratt's Theory of Constraints (TOC): lessons from the international literature
Steven J. Balderstone & Victoria J. Mabin, Victoria University of Wellington, conference paper, 1998
https://www.tocinstitute.org/uploads/1/2/7/9/12796657/toc_impact_study.pdf

The performance of the theory of constraints methodology: Analysis and discussion of successful TOC applications
Victoria J. Mabin & Steven J. Balderstone, International Journal of Operations & Production Management 23(6), 2003 · via Panizzolo, 2016
https://doi.org/10.1108/01443570310476636

Theory of Constraints (TOC) Production and Manufacturing Performance
Roberto Panizzolo, International Journal of Industrial Engineering and Management 7(1), 2016
https://www.research.unipd.it/retrieve/e14fb26b-f0bc-3de1-e053-1705fe0ac030/ijiem_vol7_no1_3.pdf

The Five Focusing Steps (POOGI)
Theory of Constraints Institute, 2021
https://www.tocinstitute.org/five-focusing-steps.html

Outcomes of managing healthcare services using the Theory of Constraints: A systematic review
Gabriel M. Bacelar-Silva, James F. Cox III &

EDITORIAL METHOD

Each figure was checked for its year, its publisher and what exactly it measures. Where the publisher's page could not be opened, the figure was checked against independent summaries and is marked “via”. The editors' interpretation is marked “Our reading”. Figures that could not be confirmed are not in the issue.



MANAGEMENT REVIEW

Management without theatre.

Every issue, one management question, checked against the best research.

ALL ISSUES

stivencatalyst.com/magazine/management-review.html